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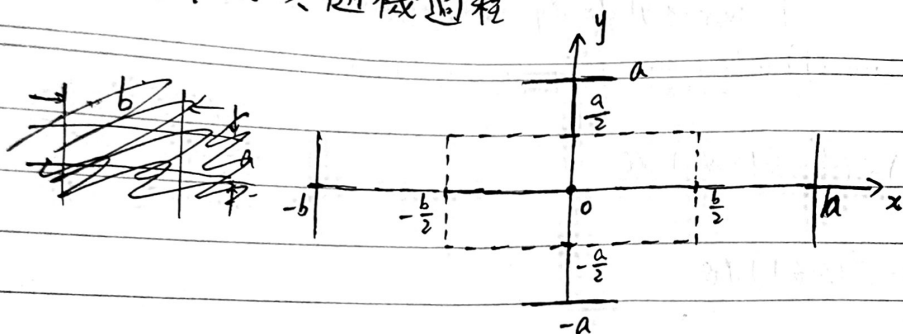
年級: 微 91

科目: 概率論與隨機過程

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Without the loss of generality, denote one of the intersection point is 0
generate the xOy plane, x -axis is parallel to horizontal line.

Denote $(X, Y) \in [-\frac{l}{2}, \frac{l}{2}] \times [-\frac{a}{2}, \frac{a}{2}]$ is the midpoint of the needle.

θ is the angle, the needle and x -axis ($\theta \in [0, 2\pi)$)

$$f_{X,Y}(x,y) = \frac{1}{ab}, \quad f_{\theta}(\theta) = \frac{1}{2\pi}$$

Denote N is the number of rectangle sides crossed by the needle.

$$P_N(2) = P(|X| \leq \frac{l}{2} |\cos \theta|, |Y| \leq \frac{l}{2} |\sin \theta|)$$

$$\begin{aligned} &= \int_0^{2\pi} \int_{-\frac{l}{2} |\cos \theta|}^{\frac{l}{2} |\cos \theta|} \int_{-\frac{l}{2} |\sin \theta|}^{\frac{l}{2} |\sin \theta|} f_{X,Y}(x,y) dx dy \cdot f_{\theta}(\theta) d\theta \\ &= \int_0^{2\pi} \frac{1}{2\pi ab} \cdot l^2 |\sin \theta \cos \theta| d\theta \\ &= \frac{l^2}{4\pi ab} \int_0^{2\pi} |\sin 2\theta| d\theta \\ &= \frac{l^2}{2\pi ab} \int_0^{\pi} |\sin 2\theta| d\theta \\ &= \frac{l^2}{\pi ab} \int_0^{\frac{\pi}{2}} \sin 2\theta d\theta \\ &= \frac{l^2}{2\pi ab} \times 2 = \frac{l^2}{\pi ab} \end{aligned}$$

$$P_N(0) = P(|X| > \frac{l}{2} |\cos \theta|, |Y| > \frac{l}{2} |\sin \theta|)$$

$$= \int_0^{2\pi} f_{\theta}(\theta) d\theta \cdot \int_{\left([-\frac{l}{2}, \frac{l}{2}] \times [-\frac{a}{2}, \frac{a}{2}]\right) \setminus \left([-\frac{l}{2} |\cos \theta|, \frac{l}{2} |\cos \theta|] \times [-\frac{l}{2} |\sin \theta|, \frac{l}{2} |\sin \theta|]\right)} f_{X,Y}(x,y) dx dy$$

$$\begin{aligned}
 P_N(0) &= P\left(|X| > \frac{l}{2}|\cos \theta|, |Y| > \frac{l}{2}|\sin \theta|\right) \\
 &= \int_0^{2\pi} f_{\theta}(\theta) \int_{\left(-\frac{l}{2}, \frac{l}{2}\right) \setminus \left[-\frac{l}{2}|\cos \theta|, \frac{l}{2}|\cos \theta|\right] \times \left(-\frac{l}{2}, \frac{l}{2}\right) \setminus \left[-\frac{l}{2}|\sin \theta|, \frac{l}{2}|\sin \theta|\right]} f_{X,Y}(x,y) dx dy \\
 &= \int_0^{2\pi} \frac{1}{2\pi ab} \times 4 \times \left(\frac{l}{2} - \frac{l}{2}|\cos \theta|\right) \times \left(\frac{l}{2} - \frac{l}{2}|\sin \theta|\right) d\theta \\
 &= \int_0^{2\pi} \frac{1}{2\pi ab} (b - l|\cos \theta|)(a - l|\sin \theta|) d\theta \\
 &= \frac{1}{2\pi ab} \int_0^{2\pi} [ab - l(a|\cos \theta| + b|\sin \theta|) + l^2|\sin \theta \cos \theta|] d\theta \\
 &= 1 + \frac{l^2}{\pi ab} - \frac{2l}{\pi ab} \int_0^{\frac{\pi}{2}} (a \cos \theta + b \sin \theta) d\theta \\
 &= 1 + \frac{l^2}{\pi ab} - \frac{2l(a+b)}{\pi ab}
 \end{aligned}$$

$$P_N(1) = 1 - P_N(0) - P_N(2) = \frac{2l(a+b)}{\pi ab} - \frac{2l^2}{\pi ab} = \frac{2l(a+b-l)}{\pi ab}$$

$$P_N(n) = \begin{cases} 1 + \frac{l^2}{\pi ab} - \frac{2l(a+b)}{\pi ab}, & n=0 \\ \frac{2l(a+b-l)}{\pi ab}, & n=1 \\ \frac{l^2}{\pi ab}, & n=2 \\ 0, & \text{otherwise} \end{cases}$$

$$E[N] = 1 \times \frac{2l(a+b-l)}{\pi ab} + 2 \times \frac{l^2}{\pi ab} = \frac{2l(a+b)}{\pi ab}$$

$$P(N \geq 1) = 1 - P_N(0) = \frac{l(2a+2b-l)}{\pi ab}$$

2. Denote the appointment duration of two students are X and Y minutes.

$$f_X(x) = \lambda e^{-\lambda x}, f_Y(y) = \lambda e^{-\lambda y}, \text{ the 1st student } z_0 = E[X] = E[Y] = \frac{1}{\lambda} \Rightarrow \lambda = \frac{1}{30}$$

Denote the time between the arrival of 1st student and the departure of 2nd student is T

$$T(X, Y) = \begin{cases} \frac{5}{2} + Y, & X \leq 5 \\ X + Y, & X > 5 \end{cases}$$

$$E[T] = \int_0^{\infty} \int_0^{\infty} T(x,y) f_X(x) f_Y(y) dx dy = \int_0^{\infty} \int_0^{\infty} T(x,y) \lambda^2 e^{-\lambda(x+y)} dx dy$$

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$$E[T] = \int_0^{\infty} \int_0^5 (y+5) \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda y} dy dx + \int_0^{\infty} \int_5^{\infty} (x+y) \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda y} dy dx$$

$$= \int_0^{\infty} \int_0^{\infty} (x+y) \cdot \lambda^2 e^{-\lambda(x+y)} dx dy + \int_0^{\infty} \int_5^5 (5-x) \lambda^2 e^{-\lambda x} \cdot e^{-\lambda y} dy dx$$

$$= \int_0^{\infty} t^2 \lambda^2 e^{-\lambda t} dt + \int_0^5 (5-x) \lambda^2 e^{-\lambda x} dx$$

$$= \lambda \int_0^{\infty} t^2 \lambda e^{-\lambda t} dt + \int_0^5 x \lambda e^{-\lambda(5-x)} dx$$

$$= \frac{2}{\lambda^2} \cdot \lambda + e^{-5\lambda} \int_0^5 x e^{\lambda x} dx$$

$$= \frac{2}{\lambda} + \frac{e^{-5\lambda}}{\lambda^2} \left(5e^{5\lambda} - \int_0^5 e^{\lambda x} dx \right) = \frac{2}{\lambda} + \frac{5}{\lambda^2} - \frac{e^{-5\lambda}}{\lambda^2} (e^{5\lambda} - 1)$$

$$= \frac{2}{\lambda} + \frac{5}{\lambda^2} - \frac{1}{\lambda^2} + \frac{e^{-5\lambda}}{\lambda^2}$$

$$= \frac{7}{\lambda} - \frac{1}{\lambda^2} + \frac{e^{-5\lambda}}{\lambda^2} = 7 \times 30 - (30)^2 + e^{-\frac{1}{6}} \times (30)^2$$

$$= 210 - (1 - e^{-\frac{1}{6}}) \times 900$$

$$= 450 = (1 - e^{-\frac{1}{6}}) \times 2700$$

$$= 2 \times 30 + 5 - (1 - e^{-\frac{1}{6}}) \times 900$$

$$= 65 = (1 - e^{-\frac{1}{6}}) \times 900$$

$$= \frac{2}{\lambda} + 5 - \frac{1}{\lambda} (1 - e^{-5\lambda}) = \frac{1}{\lambda} (1 + e^{-5\lambda}) + 5 = 30 \times (1 + e^{-\frac{1}{6}}) + 5 \text{ min}$$

$$3. (a) E[P] = \int_0^1 p^2 e^p dp = e^{-2} \int_0^1 p e^p dp = e^{-2} \times (e - \int_0^1 e^p dp) = e^{-2} \times (e - e + 1) = e^{-2}$$

$$(b) f_P | \text{a coin toss resulted in heads} (P) = \begin{cases} \frac{Pe^P}{e-2}, & P \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

$$(c) P(\text{2nd head} | \text{1st head}) = \frac{\int_0^1 p^2 e^p dp}{\int_0^1 p e^p dp} = \frac{\frac{1}{8} \int_0^1 t^2 e^t dt}{e-2} = \frac{4e^2 - 2 \int_0^1 t e^t dt}{8(e-2)}$$

$$= \frac{2e^2 - (2e^2 - (4e^2 - 1))}{8(e-2)} = \frac{e^2 - 1}{8(e-2)}$$

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4. Let l is the length of the rod.

Denote X is the longer part of the broken rod in first time

Y is the length of the longer part of the broken rod which length is X .

$$f_X(x) = \begin{cases} \frac{2}{l}, & x \in [\frac{l}{2}, l) \\ 0, & \text{otherwise} \end{cases}, \quad f_{Y|X}(y|x) = \begin{cases} \frac{2}{x}, & y \in [\frac{x}{2}, x) \\ 0, & \text{otherwise} \end{cases}$$

$$f_{X,Y}(x,y) = \frac{4}{lx}, \quad \text{the length of 3 sides are } Y, X-Y, l-X$$

~~given that~~

$Y, X-Y, l-X$ can form triangle, $X \leq 2Y < l \leq 2X$

if and only if

$$\begin{aligned} \text{(a)} \quad P(\Delta) &= \int_{\frac{l}{2}}^l \int_{\frac{x}{2}}^{\frac{l}{2}} f_{X,Y}(x,y) dx dy = \int_{\frac{l}{2}}^l \frac{4}{lx} \cdot \frac{l-x}{2} dx = \int_{\frac{l}{2}}^l 2 \left(\frac{1}{x} - \frac{1}{l} \right) dx \\ &= 2 \ln x \Big|_{\frac{l}{2}}^l - \frac{2}{l} \times \frac{l}{2} = 2 \ln 2 - 1 = \ln \left(\frac{4}{e} \right) \end{aligned}$$

(b) if the longer part is chosen, ~~the~~ 3 sides are $\frac{X}{2}, \frac{X}{2}, l-X$

if the shorter part is chosen, 3 sides are $X, \frac{l-X}{2}, \frac{l-X}{2}$

Obvious, $X \geq \frac{l-X}{2} + \frac{l-X}{2} \Leftrightarrow X, \frac{l-X}{2}, \frac{l-X}{2}$ cannot form a triangle.

$\frac{X}{2}, \frac{X}{2}, l-X$ can form a triangle.

Therefore, $P(\Delta) = P(\text{the longer part is chosen}) = \frac{1}{2}$

$$\text{(c)} \quad P(\text{the triangle is obtuse}) = \cancel{P\left(\frac{l}{2} < X < l-X\right)} = \cancel{P\left(X < 2(\sqrt{2}-1)l\right)}$$

$$= \cancel{P\left(\frac{l}{2} \leq X \leq \frac{1}{\sqrt{2}}(l-X)\right)}$$

$$= P\left(\frac{\sqrt{2}l}{4} \leq \frac{\sqrt{2}}{2}X \leq l-X\right) = P\left(\frac{l}{2} \leq X < \frac{(2-\sqrt{2})}{2}l\right)$$

$$= \frac{2}{l} \cdot \left[\frac{2-\sqrt{2}}{2} - \frac{1}{2} \right] l = \cancel{4-2\sqrt{2}} \quad 3-2\sqrt{2}$$